

Aufg. 3 PI-Zustandsregelung mit Butterworth-Verhalten ($n=2$)

Grenzkreisfrequenz: ω_{GR} ; $s_{n2} = -\omega_{GR}$

$$\dot{x} = \underbrace{\begin{bmatrix} 0 & \frac{1}{J} \\ 0 & -\frac{1}{T_{EM}} \end{bmatrix}}_A \cdot x + \underbrace{\begin{bmatrix} 0 \\ \frac{1}{T_{EM}} \end{bmatrix}}_B \cdot y + \underbrace{\begin{bmatrix} -\frac{1}{J} \\ 0 \end{bmatrix}}_{Bz} z ; \quad v = \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_C \cdot x$$

mit $x = \begin{bmatrix} \omega \\ M \end{bmatrix}$; $y = M^*$; $z = M_L$; $v = \omega$

Zustandsrückführung, Koeff. PI-Regler, $Q(s)$?

1. auf RNF transformieren:

$$\underline{S}_u = [B, AB] = \begin{bmatrix} 0 & \frac{1}{J \cdot T_{EM}} \\ \frac{1}{T_{EM}} & -\frac{1}{T_{EM}} \end{bmatrix} \Rightarrow \det \underline{S}_u = -\frac{1}{J T_{EM}^2} \neq 0 \Rightarrow \text{steuerbar}$$

$$\Rightarrow \underline{S}_u^{-1} = -J T_{EM}^2 \begin{bmatrix} -\frac{1}{T_{EM}} & -\frac{1}{J T_{EM}} \\ -\frac{1}{T_{EM}} & 0 \end{bmatrix} = \begin{bmatrix} J & T_{EM} \\ J T_{EM} & 0 \end{bmatrix} \Rightarrow \underline{T}_R = \begin{bmatrix} J T_{EM} & 0 \\ 0 & T_{EM} \end{bmatrix}$$

$$\Rightarrow \underline{T}_R^{-1} = \frac{1}{J T_{EM}^2} \begin{bmatrix} T_{EM} & 0 \\ 0 & J T_{EM} \end{bmatrix} = \begin{bmatrix} \frac{1}{J T_{EM}} & 0 \\ 0 & \frac{1}{T_{EM}} \end{bmatrix}$$

$$\Rightarrow \underline{A}_R = \underline{T}_R \underline{A} \underline{T}_R^{-1} = \begin{bmatrix} 0 & T_{EM} \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{J T_{EM}} & 0 \\ 0 & \frac{1}{T_{EM}} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -\frac{1}{T_{EM}} \end{bmatrix}$$

$$\underline{B}_R = \underline{T}_R \cdot B = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \underline{C}_R = C \underline{T}_R^{-1} = \begin{bmatrix} \frac{1}{J T_{EM}} & 0 \end{bmatrix}$$

2. Polynom wdg. / Koeff. best. $a_n = a_2 = 1$

$$\tilde{P}_R = P_R \cdot (s - s_{n2}) = (s^2 + \sqrt{2} \omega_{GR} \cdot s + \omega_{GR}^2) \cdot (s + \omega_{GR}) = s^3 + s^2 (\sqrt{2} \omega_{GR} + \omega_{GR}) + s (\omega_{GR}^2 + \sqrt{2} \omega_{GR}^2) + \omega_{GR}^3$$

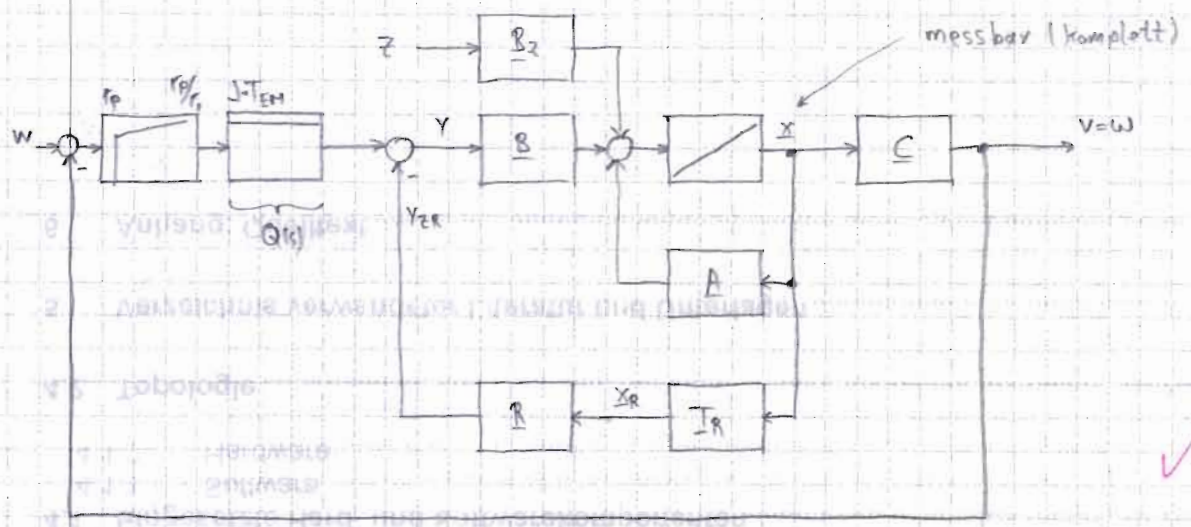
$$r_s = \frac{\tilde{p}_0}{\tilde{p}_3} a_2 = \omega_{GR}^2 ; \quad r_p = -\frac{r_s}{s_{n2}} = -\frac{\omega_{GR}^2}{-\omega_{GR}} = \omega_{GR}$$

$$r_1 = \tilde{p}_1 + \frac{r_s}{s_{n2}} - a_0 = \sqrt{2} \omega_{GR}^2$$

$$r_2 = \tilde{p}_2 - a_1 = \omega_{GR} (1 + \sqrt{2}) - \frac{1}{T_{EM}}$$

$$z_s(s) = b_0 + b_1 \cdot s = \frac{1}{J T_{EM}} \Rightarrow Q(s) = 1/z_s(s) = J T_{EM}$$

Aufg.3 (Fortsetzung)



Aufg. 4

(a) Regel:

$$G_R(z) = K_P$$

Strecke:

$$T_t = T$$

$$G_S(s) = \frac{1-e^{-sT}}{s} \cdot e^{-sT_t} \cdot \frac{1}{s} \Rightarrow G_S(z) = (1-z^{-1})z^{-1} \cdot Z\left\{\frac{1}{s^2}\right\}$$

$$G_0(z) = G_R(z) \cdot G_S(z) = \frac{K_P T}{z(z-1)} = \frac{z_0}{N_0}$$

$$G_1(z) = (1-z^{-1})z^{-1} \cdot T \frac{z}{(z-1)^2}$$

char. Polynom:

$$= \frac{z-1}{z} z^{-1} \cdot T \frac{z}{(z-1)^2}$$

$$N(t) = z_0 + N_0$$

$$G_S(z) = \frac{T}{z(z-1)}$$

$$= K_P T - z + z^2$$

$$z_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

aperiodisch, wenn " $\sqrt{\quad}$ "

$$\Rightarrow b^2 = 4ac \Rightarrow 1 = 4 \cdot K_P T \Rightarrow K_P = \frac{1}{4T}$$

